

Power BI Capstone Project 3

Merchandising & Product Profitability — Contoso Retail Group

Data file: Contoso Retail Data.xlsx • Reference: Data Dictionary (Cheat Sheet).pdf • Deliverable: one .pbix file with a two-page report

1. The data

The workbook Contoso Retail Data.xlsx contains eight sheets. Note that there are **two fact tables** in this extract, not one:

Sheet	What it holds	Approx. rows
sales	One row per product line sold — the main fact table	121,000
returns	One row per line that came back — the second fact table	9,900
product	The product catalogue: category, colour, weight, cost, price	2,500
manufacturer	The eleven brands, their tier and where they are supplied from	11
store	The 73 stores: location, size, opening and closing dates, status	73
geography	Countries, states and regions	595
promotion	Marketing campaigns and the discount each one gave	65
currency	The five trading currencies	5

4. Task 1 — Prepare the data in Power Query

The extract contains a number of data quality problems. **You are not told what they are.** Finding them is part of the assessment. Work through every table and fix everything you find, using Power Query only — **do not edit the Excel file**

Profile every table before you transform it

In the Power Query Editor, turn on **View > Column quality**, **Column distribution** and **Column profile**, and set the profiler to **Column profiling based on entire data set** (bottom status bar). Then check, for every table:

- **Data types** — is every column the type it ought to be? A text icon on a date, a number, or a percentage is a warning sign.
- **Distinct values** — use *Group By* on any column you plan to slice or chart by. If a column with five real values returns twenty-one groups, something is wrong. Watch out for the difference between a value written in a different *case* and a value written in different *words* — one is fixed by formatting, the other has to be replaced by hand.
- **Duplicates** — every dimension table must have a unique key column. Check each one.
- **Columns that carry no information** — *Column distribution* will show you any column with a single distinct value, and *Column quality* any column that is 100% empty.

Rules for your cleaning work

- All transformations must be done in **Power Query**.

- **Rename every applied step** to something meaningful — "Removed duplicate products", not "Removed Duplicates". Your Applied Steps list is marked.
- Do not delete rows to make a problem go away unless the rows are genuinely duplicates.
- After you convert a numeric column, **check its scale**, not just its type. A number can be the right data type and still be a hundred times too big.
- When you finish, write a short **Cleaning Log** (half a page, submitted separately) listing each problem you found, the table it was in, and how you fixed it.

5. Task 2 — Build the data model

5.1 Create a Date table

There is no calendar in the source data. Create one with DAX covering the full period of the data — remember that returns continue for a few weeks after the last order. It must contain at least: Date, Year, Quarter, Month Number, Month Name.

5.2 Two fact tables, shared dimensions

This is the part of the project that is different from the others, so read it carefully.

sales and returns are both fact tables, and they sit at different grains — one row per line sold, one row per line returned. **Do not create a relationship between them.** Instead, connect each of them separately to the dimensions they share:

- Date joins sales[OrderDate] **and** returns[ReturnDate] — two separate relationships, both active, because they go to different tables.
- store joins both fact tables.
- product joins both fact tables.

If you find yourself trying to join returns to sales on OrderKey and LineNumber, stop. Power BI cannot make a relationship on two columns at once, and you do not need one. Every question in this brief can be answered through the shared dimensions.

5.3 The rest of the model

manufacturer joins to product, not to the facts. geography and currency join to store. promotion joins to sales only. Every relationship must be **one-to-many** with a **single** cross-filter direction.

6. Task 3 — Create the measures

Create the following as **measures** (not calculated columns), in a dedicated measures table or folder. Each is defined here in business terms — working out the DAX is your task.

#	Measure	What it must return
1	Total Revenue	Sales revenue after discount, converted to US dollars
2	Total Cost	Cost of the goods sold, converted to US dollars
3	Gross Profit	Total Revenue minus Total Cost
4	Gross Margin %	Gross Profit as a share of Total Revenue
5	Total Units	The number of units sold

#	Measure	What it must return
6	Average Selling Price	Total Revenue divided by Total Units
7	Gross Sales	What the goods sold would have earned at full list price, before any discount in US dollars . The standard retail term for revenue before deductions
8	Discount Given	Gross Sales minus Total Revenue — the money handed back to customers. Retailers also call this the <i>markdown</i>
9	Discount Depth %	Discount Given as a share of Gross Sales — how deep the average discount cuts
10	% Revenue on Promotion	The share of revenue that came from lines carrying any discount
11	Returned Units	The number of units sent back
12	Returned Value	What those units had been sold for, in US dollars
13	Return Rate %	Returned Units as a share of Total Units — this one reads from both fact tables
14	Net Revenue	Total Revenue minus Returned Value
15	Revenue YoY %	Growth of Total Revenue against the same period last year

Read business rule 1 in the cheat sheet before you write measures 1, 2, 7 and 12. Every amount in both fact tables is recorded in the store's own currency. A measure that adds them up without converting will produce a large, plausible-looking, and completely wrong number.

The revenue ladder

Three of your measures are the same money viewed at three points, and it helps to see them as one ladder rather than three unrelated numbers:

```

Gross Sales           everything that left the shelf, at full ticket price
  less Discount Given
= Total Revenue       what the customer was actually charged
  less Returned Value
= Net Revenue         what Contoso kept
  
```

Every level answers a different person's question. Merchandising cares about Gross Sales because it is what the range was *capable* of earning. Finance reports Total Revenue. Operations lives with Net Revenue. When you put a number on a visual, be clear which rung you are standing on and say so in the title.

Calculated columns

Create these two as **calculated columns**:

- `product[Initial Margin %]` — the margin built into the shelf price, before any discount or return. Retail calls this the *initial* (or *intake*) margin, to distinguish it from the *achieved* margin you actually banked once the discounts came off.
- `product[Weight in Grams]` — the weight converted to a single unit so it can be summed. See business rule 2 in the cheat sheet.

7. Task 4 — Build the report

Two pages and every visual must have a meaningful title.

Page 1 — "Product Profitability"

#	Visual	What it must show
1	Five KPI cards	Total Revenue, Gross Profit, Gross Margin %, Total Units, Average Selling Price
2	Clustered bar chart	The top 10 subcategories by Gross Profit
3	Scatter chart	One point per category: Gross Margin % on one axis, Return Rate % on the other, bubble size by revenue. This one chart answers more of the brief than any other on the page
4	Matrix	Brand tier down the rows, showing Total Revenue, Gross Margin %, Return Rate % and Net Revenue, with conditional formatting on the margin column
5	Bar chart	Revenue by manufacturer, sorted highest to lowest
6	Three slicers	Year, Category, and Brand Tier

Page 2 — "Discounting & Returns"

#	Visual	What it must show
1	Five KPI cards	Discount Given, Discount Depth %, Returned Value, Return Rate %, Net Revenue
2	Clustered column chart	Gross Margin % by promotion type, so the cost of each kind of campaign is visible side by side
3	Bar chart	The top 10 subcategories by Returned Value
4	Donut chart	Returned units split by return reason — five slices, which is the most a donut should carry
5	Line and clustered column chart	Total Revenue as columns and Gross Margin % as a line, with the discount rate on the axis — this shows directly what each level of discount costs
6	Three slicers	Year, Manufacturer, and Promotion Type

Report standards

- One consistent colour theme across both pages. Do not use the default Power BI palette unchanged.
- Large numbers displayed in thousands or millions, not raw digits. Percentages to one decimal place, except Discount Depth % which needs two.
- Slicers grouped together in the same position on both pages.
- Write meaningful clear title for each visual

three rules that will be marked:

1. **No pie or donut with more than five slices.**
2. **No stacked chart with more than four or five series.**
3. **Two series in the same unit share one Y axis.** If you put Total Revenue on the left axis and Revenue LY on the right, Power BI scales them independently and the lines will appear to cross where they do not. Force both onto the primary axis.

10. How you will be marked

Area	What we are looking for	Marks
Data cleaning	Every quality issue found and fixed in Power Query; steps clearly renamed.	30
Data model	Correct model with two fact tables and shared dimensions; Date table created; both date relationships present; no relationship attempted between the two facts; all relationships one-to-many, single direction.	10
Measures	All 15 measures present and returning correct values; currency handled correctly; both calculated columns created.	30
Report & design	All required visuals present and correctly configured; consistent theme; readable formatting; slicers working.	30
Total		100